

A stable and explicit fluid–structure interaction solver based on lattice-Boltzmann and immersed boundary methods

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ABSTRACT

Fluid–structure interaction (FSI) occurs in a wide range of contexts, from aeronautics to biological systems. To numerically address this challenging type of problem, various methods have been proposed, particularly using implicit coupling when the fluid and the solid have the same density, i.e., the density ratio is equal to 1. Aiming for a computationally efficient approach capable of handling strongly coupled dynamics and/or realistic conditions, we present an alternative to the implicit formulation by employing a fully explicit algorithm. The Lattice Boltzmann Method (LBM) is used for the fluid, with the finite element method (FEM) utilized for the structure. The Immersed Boundary Method (IBM) is applied to simulate moving and deforming boundaries immersed in fluid flows. The novelty of this work lies in the combination of Laplacian smoothing at the fluid/solid interface, an improved collision model for the LBM, and a reduction of non-physical frequencies on the structure mesh. The use of these adaptations results in a solver with remarkable stability properties, a primary concern when dealing with explicit coupling. We validate the numerical framework on several challenging test cases of increasing complexity, including 2D and 3D configurations, density ratio of 1, and turbulent conditions.

1. Introduction

Fluid structure interaction (FSI) is a ubiquitous phenomenon in nature and industry, for instance in immersed risers for offshore applications, the deflection of airplane wings, the blood flow applying pressure on the heart's valves, or wind turbines [1–4]. With the development of numerical methods and high-performance computing over the last decades, the simulation of large-scale three-dimensional FSI problems is nowadays reachable in reasonable computational times. To do so, a classical approach consists of solving the Navier Stokes equations for the fluid and to use a finite element method (FEM) for the solid, both domains being coupled by the Arbitrary Lagrangian–Eulerian method [5–7]. Although this framework is widely used in commercial software, it requires re-meshing of the fluid domain when it becomes too distorted, which is time consuming and can be limiting in the case of large deformations. On the other hand, the Immersed Boundary Method (IBM), introduced in 1973 by [8], has the advantage to require no re-meshing nor coinciding meshes between the fluid and solid domains. This method is by construction well adapted to fixed cartesian meshes, and is thus easy to couple with the cartesian lattice framework of lattice Boltzmann method (LBM) [9]. Recent works in the literature are coupling LBM, IBM and FEM included [10–18], and this is the strategy also adopted in the present work.

When dealing with FSI cases, the density ratio between the solid and the fluid ρ_s/ρ_f is crucial for the stability of the solver [19]. This is linked to the relative influence of the fluid on the solid and *vice-versa*. For instance, in the case of flows in the human body,

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the density ratio typically tends to one, as most human tissues are composed of water with a density similar to the blood density. Thus, in this case, both fluid and solid interact with the same magnitude, same characteristic times, leading to a strong coupling, also named high-added mass effect.

From a numerical point of view, a coupling algorithm is required to integrate in time the fluid and structure fields, under the condition of the continuity of solutions (pressure, displacement, velocity) at the interface. Depending on the weak or strong nature of the coupled dynamics, explicit or implicit algorithms are used. First, explicit approaches are easy to implement, straightforward to integrate in time due to a staggered formulation, but limited to small-time steps and low added mass effects ($\rho_s/\rho_f \neq 1$) because of the time lag between the solvers [20,21], which leads to an imbalance of energy at the fluid/structure interface [16,22]. The second family of algorithms, mainly used in the case of high-added mass effect, are implicit, with very low or even no restriction on the time step, nor on the density ratio. Two variants exist to enforce the continuity of macroscopic quantities at the interface:

- The first one is the implicit staggered approach which checks the continuity at the interface at the end of the coupling time step [23–25]. If the continuity is respected, the algorithm performs the next time step; otherwise it sub-iterates with the last solution until reaching the level of continuity desired by the user. This approach allows for keeping the modularity of the two solvers and does not require any heavy modification of the original codes. Nevertheless, the process of sub-iteration increases the computational time and does not always guarantee convergence [26].
- The second variant is the implicit monolithic [27–29] approach, which formulates the fluid and the solid equations in a single system to solve in one iteration. Therefore, both the structure and the fluid solutions are obtained at the same time and the mathematical formulation is exact. Although there is no stability problem, this formulation can be restrictive and not modular. Indeed, as the monolithic solver requires to write the fluid and solid equations already coupled in one system, each upgrade of the fluid or solid part (i.e. the implementation of turbulence, or hyperelastic law, or contact between bodies, etc.) needs to be added to the system leading to heavy and nonlinear matrices to solve, whereas in the case of a staggered algorithm each feature of the solver does not add complexity to the coupling part.

The use of explicit coupling is limited in terms of physics, while implicit coupling can be cost prohibitive at large scales. To address this, research works have combined the efficiency and accuracy of both methods using semi-implicit coupling [22,30]. Additionally, other works aimed stabilizing the explicit resolution directly, such as in [31], by subtracting the added mass component from the fluid forcing. In LBM, [32] employed a sub-step strategy in the solid solver and a Lagrangian force decomposition for the IBM with feedback based on parameters.

In this work, with a focus on large-scale FSI cases and the aim of maximizing computational efficiency and modularity, we adopt an explicit coupling approach. The advantage and novelty of the present work is that it does not need to subtract the physical added mass effect from the fluid nor to use sub-iterations of the structure solver and is only limited by the fluid time step. To obtain this new result, three key features are implemented to maintain stability: The HRR collision model for the Lattice Boltzmann solver [33], a Laplacian smoothing of the force at the interface, and a selective mass scaling for the Finite Element solver [34].

The plan of this article is as follows. In the next Section 2, we start with an overview of the LBM, IBM and FEM approaches independently. Then in Section 3 the coupling section presents the mathematical formulation with the benefits of the three stabilization tools and the numerical challenges related to FSI. To validate this fully explicit approach, we perform in Section 4 six test cases and we measure the accuracy and stability of the code. It is worth pointing out that most of these cases are performed in literature using an implicit coupling to obtain stable solutions. Finally, in the conclusion in Section 5, we summarize the findings and put forward the good properties of our approach to capture complex physics and deal with large-scale configurations in an efficient way.

2. Numerical methodology

2.1. Lattice Boltzmann method

A brief overview of the Lattice-Boltzmann Method (LBM) is presented here. For more information about the LBM, the reader can refer to [35].

The LBM considers the fluid at a mesoscopic scale, as groups of molecules moving in function of their natural streaming and collision with other groups of molecules. A group of molecules is described by the function $f(\mathbf{x}, \mathbf{c}, t)$, named “particle distribution function”, which gives the probability density of particles to be at position $\mathbf{x}$, with velocity $\mathbf{c}$ at time t . The LBM consists of solving Eq. (1) to determine f , and then use f to reconstruct the macroscopic flow quantities (density ρ , momentum ρv_α , and momentum flux tensor $\Pi_{\alpha\beta}$). The Greek letters α and β represent the cartesian coordinates.

$$\frac{\partial f}{\partial t} + \mathbf{c} \cdot \nabla f = \Psi(f) \quad (1)$$

The left-hand side of Eq. (1) is the temporal evolution of f with an advection term for the particle spreading. The right side contains the collision step Ψ . After discretization in space, time, and velocities, on a so-called D3Q19 lattice (Fig. 1) as it is classically well suited to simulate athermal weakly compressible flows, the Lattice-Boltzmann equation is given by Eq. (2):

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = f_i^{eq}(\mathbf{x}, t) + \left(1 - \frac{\Delta t}{\tau}\right) f_i^{neq}(\mathbf{x}, t) + \frac{\Delta t}{2} h_i(\mathbf{x}, t) \quad (2)$$

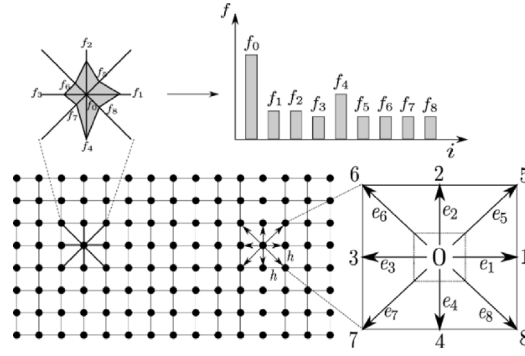


Fig. 1. Representation of a D2Q9 LBM discretization (for the sake of simplicity) with a representation of the density of probability (function f) for a group of particles to be at a specific fluid node with one of the nine discretizes velocities at time t .

The term τ represents the relaxation rate between particle collisions. The equilibrium function f_i^{eq} represents the local kinetic equilibrium distribution and is written by:

$$f_i^{eq} = \omega_i \left[\rho + \frac{c_{i\alpha} \rho v_\alpha}{c_s^2} + \frac{a_{\alpha\beta}^{(2),eq} H_{i\alpha\beta}^{(2)}}{2c_s^4} + \frac{a_{\alpha\beta\gamma}^{(3),eq} H_{i\alpha\beta\gamma}^{(3)}}{6c_s^6} \right] \quad (3)$$

Where $H_{i\alpha\beta}$ and $H_{i\alpha\beta\gamma}$ are respectively the second and third order Hermite polynomials, ω_i is the weight coefficient specific to each velocity discretization and c_s is the sound velocity in the fluid.

The non-equilibrium distribution function f_i^{neq} Eq. (4) is also expanded in Hermite polynomials and reconstructed by a Recursive Regularized BGK LBM [33] :

$$f_i^{neq} = \omega_i \left[\frac{a_{\alpha\beta}^{(2),neq} H_{i\alpha\beta}^{(2)}}{2c_s^4} + \frac{a_{\alpha\beta\gamma}^{(3),neq} H_{i\alpha\beta\gamma}^{(3)}}{6c_s^6} \right] \quad (4)$$

The non-equilibrium coefficients $a_{\alpha\beta}^{(2),neq}$ and $a_{\alpha\beta\gamma}^{(3),neq}$ are evaluated with the hybrid recursive collision model introduced in [33] to enhance numerical stability.

The HRR collision model [33] is specific to the LBM and improves the stability of the fluid solver by enforcing a traceless lattice-Boltzmann tensor (by simply removing its trace). It is observed to yield accurate and robust results for a broad range of applications and flow physics, e.g. [36–40]. It is the first time that the HRR model is used in FSI.

Using classical normalization in LBM in the weakly compressible case, the time step and spatial discretization are linked by the following relationship:

$$\Delta t = \frac{\Delta x}{c_s \sqrt{3}} \quad (5)$$

with Δx the mesh size and c_s the speed of sound in the fluid. The factor of $\sqrt{3}$ arises from the discretization of the Lattice Boltzmann Method on a regular Cartesian grid of squares, each with a dimensionless side of length $\sqrt{3}$. It is simpler to reformulate our problem so as to obtain a length of 1. Further details on this adimension process can be found in [35]. To reduce the errors due to compressibility, the following condition is used in the case of incompressible flows:

$$M_a = \frac{U}{c_s} < 0.3 \quad (6)$$

It is analog to a CFL condition. In practice, c_s acts like a parameter which can be adjusted as a function of the required time step and the velocity of the flow U , to respect the incompressible fluid assumption.

The macroscopic reconstruction is straightforward Eq. (7) as the macroscopic values correspond to the moments of f :

$$\begin{aligned} \rho &= \sum_i f_i \\ \rho v_\alpha &= \sum_i c_{i,\alpha} f_i \\ \Pi_{\alpha\beta} &= \sum_i c_{i,\alpha} c_{i,\beta} f_i \end{aligned} \quad (7)$$

In the case of an external forcing g (which will be used for FSI simulation and computed by IBM), the force term h_i is used, based on Guo's formulation [41] :

$$h_i = \rho \omega_i \left(1 - \frac{\Delta t}{2\tau} \right) \left[\frac{c_i \cdot u}{c_s^2} + \frac{c_i \cdot u}{c_s^4} c_i \right] \cdot g \quad (8)$$

and the macroscopic velocity is modified as:

$$\rho v = \sum_i c_i f_i^{eq} + \frac{\Delta t}{2} g \quad (9)$$

2.2. Immersed boundary method

An explicit direct forcing IBM is used here, completed by the correction of the no-slip boundary error [42,43], following the implementation presented and validated in [44]. To impose the boundary condition on the velocity at the solid/fluid interface, the basic idea is to use a forcing term g . A first simulation is performed without the object to obtain a predictive velocity. Then, g is computed at the interface nodes to impose the no-slip condition. In a way, it is countering the flow, which means, by the principle of reciprocal actions, that the IBM force corresponds exactly to the force applied by the fluid on the structure. Another advantage of the IBM is that a fixed mesh is used, while the motion of the boundary is simply tracked by lagrangian points defining its border. This eliminates the necessity of deforming or creating a new mesh during the calculation, both of which are being well known to be computationally expensive processes. However, as the solid mesh does not coincide with the fluid mesh, the communication between both domains requires interpolation and spreading steps.

The solid domain is denoted as Ω_s and the fluid occupies the volume Ω_f where the nodes are listed by x_j . The interface between the fluid and the object is composed of the solid nodes X_k and is defined as $\Gamma = \Omega_f \cap \Omega_s$. For the IBM algorithm, we use capital letters to indicate the variables at the lagrangian markers and lowercase letters for the eulerian nodes. Finally, we use v^* for the predicted velocity without the presence of solid.

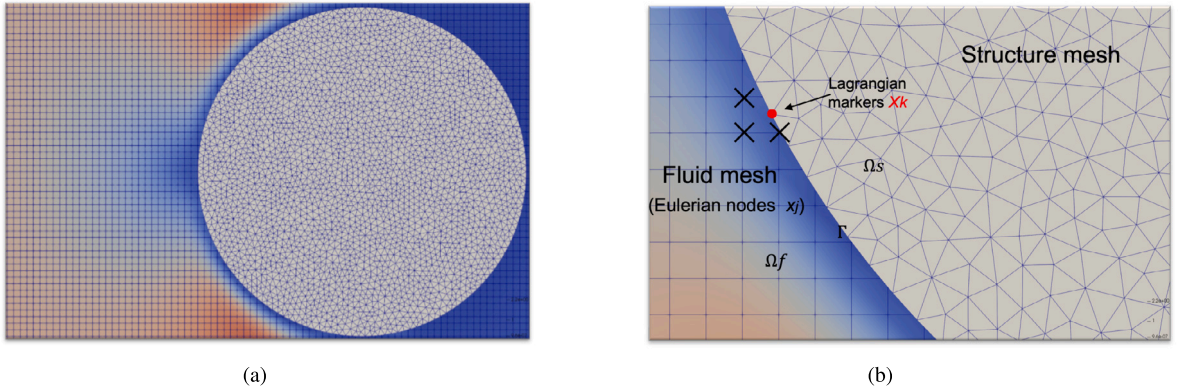


Fig. 2. (a) Disk immersed in a flow colored by the velocity, we can see the fluid sticking to the clamped solid. (b) Zoom at the interface, we can see that the nodes of the fluid and solid mesh do not coincide.

The spreading operator $S[G](x)$ of a function G is written as :

$$S[G](x) = \int_{\Gamma} G(X) \delta(x - X) dX \quad (10)$$

where dX is the infinitesimal integration element associated with the solid boundary in the continuous formulation of the integral. Its discretization is not straightforward due to an IBM-specific dimension issue explained at the end of the section. For that reason, the integration element dX is denoted as W in its discrete formulation Eq. (17). The spreading operator mathematically expresses the distribution of a given value known on a lagrangian point over the surrounding eulerian nodes.

The inverse operator, namely the interpolating operator $I[\phi](r)$ of a function ϕ on the boundary Γ , uses the same δ function given in Eq. (16). The domain Ω_f^* represents Ω_f with the addition of the fluid domain which is not visible on (Fig. 2) where the solid is immersed.

$$I[\phi](r) = \int_{\Omega_f^*} \phi(x) \delta(x - X(r)) \Delta x^3 \quad (11)$$

Following [42,43], the algorithm of the IB method is given below:

1. Update of the position of the lagrangian markers $\mathbf{X}_k$ and the velocity of the solid $\mathbf{U}^d$
2. Perform LBM without solid on the eulerian fluid nodes to obtain $\mathbf{v}^*$

Start of the loop over lagrangian nodes :

3. Interpolation of ρ and $\rho\mathbf{v}^*$ on the lagrangian markers $\mathbf{X}_k$:

$$I[\rho](\mathbf{X}_k) = \sum_j \rho(\mathbf{x}_j) \delta(\mathbf{x}_j - \mathbf{X}_k) \Delta x^3 \quad (12)$$

$$I[\rho\mathbf{v}^*](\mathbf{X}_k) = \sum_j \rho\mathbf{v}^*(\mathbf{x}_j) \delta(\mathbf{x}_j - \mathbf{X}_k) \Delta x^3 \quad (13)$$

4. Computation of the immersed boundary force :

$$\mathbf{G} = \frac{2}{\Delta t} (I[\rho]\mathbf{U}^d - I[\rho\mathbf{v}^*]) \quad (14)$$

5. Spreading of the force on the surrounding eulerian fluid nodes

$$\mathbf{g}(\mathbf{x}_j) = S[\mathbf{G}](\mathbf{x}_j) = \sum_k \mathbf{G}(\mathbf{X}_k) \delta(\mathbf{x}_j - \mathbf{X}_k) W_k \quad (15)$$

End of the loop over lagrangian nodes

6. Update of $\mathbf{v}$ Eq. (9) and f Eq. (2) in the LBM with the previously computed force

The value $\mathbf{U}^d$ represents the desired value of the fluid that we want to impose. For example, if the solid is not moving we use $\mathbf{U}^d = 0$ and $\mathbf{U}^d$ is equal to the velocity of the object in the case of solid motion. In this work, the desired velocity is given by the structure solver ($\mathbf{U}^d = \dot{\mathbf{u}}$), after solving the body motion equation. The term $\delta(\mathbf{x})$ is a smooth discretized (Fig. 3) approximation of the Dirac delta function in the interpolation and spreading operators. Its purpose is to transfer values from the lagrangian space to the eulerian space and vice versa, by respecting conservation properties on statistical moments.

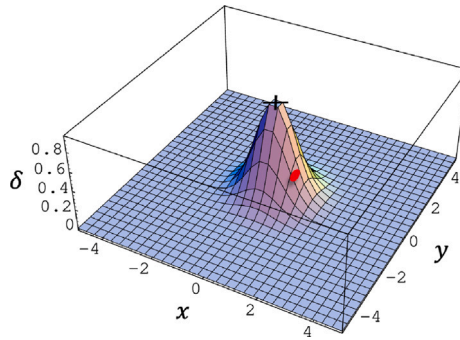


Fig. 3. Representation of the delta function in 3D with an eulerian node (black cross) and a lagrangian marker (red point). The closer they are, the higher is the delta function. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Several delta functions are used in the literature [45–47]. For the present work we use the following definition, with the ratio $d/\Delta x = 1.5$.

$$\delta(a) = \begin{cases} \frac{1}{2d} (1 + \cos(\frac{\pi a}{d})), & |a| \leq d, \\ 0, & |a| > d, \end{cases} \quad (16)$$

The spreading and interpolation operators are based on the same delta function. They are expected to be reciprocal operations, but it is not exactly the case due to discretization errors, which yield a boundary slip error and flow penetration [48]. This inaccuracy can be linked to an IBM-specific dimension issue: whereas the fluid is often specified as a volume (3 dimensions), the structure is only described through its boundary (2 dimensions). As a result, the number of dimensions in eulerian and lagrangian spaces differ, and one dimension is missing in the spreading operator. This problem has already been addressed in literature [9,49–52], with different strategies to calibrate the control volume of a Lagrangian marker W_k in order to impose the reciprocity between spreading and interpolating operators. We use here a formulation Eq. (17) that take into consideration the curvature of the boundary, making

the method applicable to a wide variety of cases. A full description of this implementation is given in [49].

$$W_k = \frac{1}{\sum_l \sum_j \delta(\mathbf{x}_j - \mathbf{X}_k) \delta(\mathbf{x}_j - \mathbf{X}_l)} \quad (17)$$

A first sum is applied on each lagrangian marker $l = 1 \dots N$, and a second sum is applied on the eulerian nodes j included in the delta function support area of point l .

To obtain a full 2-way coupled FSI solver, we then need to compute the deformation of the solid by using the IBM force, and this is done here by coupling the present framework to a finite element solid solver.

2.3. Finite element method

The finite element method (FEM) is efficient and appropriate to solve deformable structure dynamics and many efficient software based on this approach are available with an open source access. In this work, we use Calculix [53], a three-dimensional structural solver able to compute linear and non-linear cases in a static or dynamic configuration. We remind here the dynamic body motion equation Eq. (18) and the temporal discretization. The complete mathematical description of this FE solver can be found in [53].

$$[\mathbf{M}]\ddot{\mathbf{u}} + [\mathbf{C}]\dot{\mathbf{u}} + [\mathbf{K}]\mathbf{u} = \mathbf{G} \quad (18)$$

$\mathbf{M}$ is the mass matrix, $\mathbf{C}$ is the damping matrix and $\mathbf{K}$ is the stiffness matrix, $\ddot{\mathbf{u}}$, $\dot{\mathbf{u}}$ and $\mathbf{u}$ are respectively the acceleration, velocity and position of the solid nodes, and finally $\mathbf{G}$ are the fluid forces calculated by the IBM. To solve Eq. (18), for the sake of computational efficiency we choose an explicit Newmark scheme. The temporal discretization of Eq. (18) reads:

$$[\mathbf{M}]\ddot{\mathbf{u}}^{n+1} + [\mathbf{C}]\dot{\mathbf{u}}^{n+1} + [\mathbf{K}]\mathbf{u}^{n+1} = \mathbf{G}^{n+1} \quad (19)$$

The matrices $\mathbf{M}$, $\mathbf{C}$ and $\mathbf{K}$ in Eq. (19) are non-diagonal. The diagonalization is described in [54] and allows the direct inversion of the matrices for explicit resolution. The quantities $\dot{\mathbf{u}}^{n+1}$ and $\mathbf{u}^{n+1}$ can be considered as predictor values and depend on values at time t only:

$$\begin{aligned} \tilde{\mathbf{u}}^{n+1} &= \dot{\mathbf{u}}^n + (1 - \gamma)\Delta t \ddot{\mathbf{u}}^n \\ \bar{\mathbf{u}}^{n+1} &= \mathbf{u}^n + \Delta t \dot{\mathbf{u}}^n + \frac{1}{2}(\Delta t)^2(1 - 2\beta)\ddot{\mathbf{u}}^n \end{aligned} \quad (20)$$

The Newmark coefficient γ and β are respectively set to $\frac{1}{2}$ and $\frac{1}{4}$ leading to the average acceleration method second-order accurate. However this explicit scheme is not unconditionally stable.

The range of stability is limited by the propagation of the highest frequency on the structure mesh. It can be noticed that mostly numerical effects are contained in activated high frequency modes whereas real physical kinetic energy generally lies in the lower eigenfrequency domain. Thus by reducing only the upper eigenfrequency domain, the physical behavior of the case remains unaffected and the range of stability can be increased.

This treatment on the frequency is done by using a selective mass scaling [34,55] that allows increasing the solid time step and maintain the stability property.

The theoretical framework [34] is based on a modification of the original lumped mass matrix in such a way that only a subset of degrees of freedom associated with high frequencies is scaled. However, a severe scaling cannot be applied since it will introduce unwanted non-physical inertia effects in the structure. Thanks to this approach, we can use a solid time step up to ≈ 100 times higher. We would like to mention that selective mass scaling differs from standard mass scaling, which only increases the density on nodes which impact more the structural dynamics.

3. Fluid structure interaction coupling

In this section, we introduce the temporal coupling of the LBM-IBM-FEM framework. We adopt a conventional serial staggered (CSS) approach, meaning that we solve the fluid with the last position calculated by the structure solver, and we solve the deformation with the last forces calculated by the fluid solver. Therefore, each solver needs to wait for the other one to start its own time increment, as explained in (Fig. 4). The time step of the coupling is set on the time step of the fluid solver Eq. (5). If the structure solver has a lower time step, the necessary amount of mass scaling is added to fix it to the fluid time step. An explanation on the mass scaling used is given at the end of this section. This approach ensures that both solvers share the same stable time step.

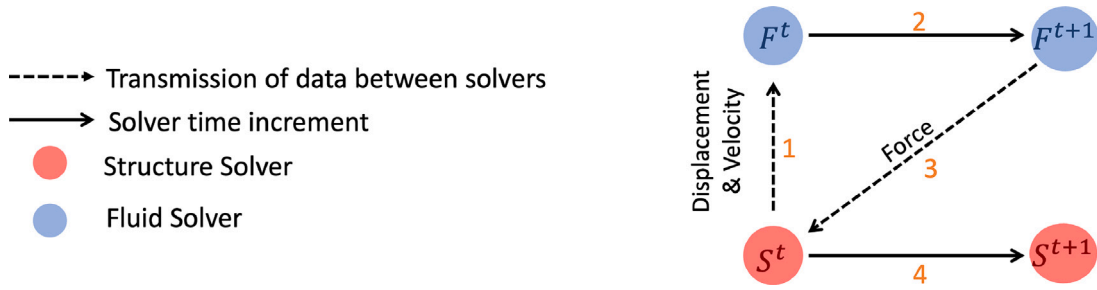


Fig. 4. Scheme of the conventional serial staggered approach (CSS).

• Both solvers are at time t .

1. The fluid solver gets the position of the structure.
2. The fluid solver solves the flow (LBM) and stores the force on the Lagrangian markers (IBM).
3. The forces are sent to the solid solver.
4. The solid solver performs its time increment (FEM).

✓ Both solvers are at time $t+1$.

The continuity of velocity and stress (σ) need to be verified at the interface of a FSI problem. In other words, fluid and solid must have the same velocity to ensure no slip condition and the force applied by the fluid on the solid has to be equal to the force of the solid to ensure the reciprocity:

$$\mathbf{v} = \frac{\partial \mathbf{u}}{\partial t} \quad \sigma_F \cdot \mathbf{n}_F = -\sigma_S \cdot \mathbf{n}_S \quad \text{at } \Gamma \quad (21)$$

Numerically, these conditions are not always easy to ensure numerically. When solving the structure dynamics by equation Eq. (19), the unknown displacement/velocity and acceleration of the object at $t+1$ is a function of the force term $\mathbf{G}$ at time $t+1$, which comes from the fluid solver:

$$\mathbf{G}^{n+1} = \frac{2}{\Delta t} (I[\rho]^{n+1} \dot{\mathbf{u}}^{n+1} - I[\rho v^*]^{n+1}) \quad (22)$$

Thus, the force term $\mathbf{G}^{n+1}$ depends on $\dot{\mathbf{u}}^{n+1}$. The fluid solver hence requires the position at $t+1$ of the object to calculate the force at $t+1$ and the solid solver requires the force at $t+1$ to calculate the position of the object at $t+1$. This condition, mandatory to ensure continuity of values at the interface, is not strictly respected using an explicit coupling, which affects the stability of the numerical coupling resulting in a first order accuracy in time [56]. Indeed, the time lag in the case of an explicit coupling leads to an under or over prediction of force, so that any small error in the solid displacement may result in large errors in the fluid pressure. To overcome this issue, we perform a Laplacian smoothing of the force over the 5 neighboring lagrangian nodes, as it can be done, even with implicit coupling [57–59]. The choice of 5 is based on the consideration that in 3D cases, the interface forms a 2D surface. Therefore, Laplacian smoothing involves the node itself, the right and left neighboring nodes, as well as the over and under nodes. In 2D cases, where the interface is a line without over and under nodes, the same Laplacian smoothing is used with also 5 nodes: the node itself and the 2 right and 2 left neighboring nodes. The stabilized CSS algorithm, validated in the next section is given in (Fig. 5).

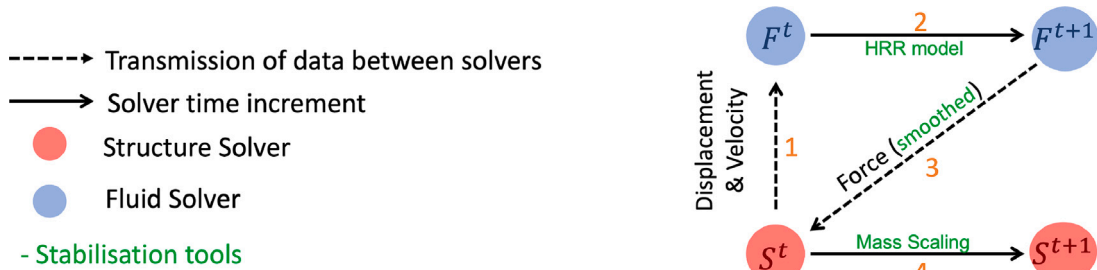


Fig. 5. Scheme of the stabilized CSS approach.

• Both solvers are at time t .

1. The fluid solver gets the position of the structure.
2. The fluid solver solves the flow (LBM), filter by the HRR model (Eq. (4)) and stores the force on the Lagrangian markers (IBM).
3. The smoothed forces are sent to the solid solver.
4. The solid solver apply a selective mass scaling performs its time increment (FEM).

✓ Both solvers are at time $t+1$.

Summary of the stabilization tools:

- **Lagrangian smoothing** : Due to the time lag, a node at the interface may perceive the applied force as either overestimated or underestimated. This discrepancy occurs locally for each Lagrangian marker, but the global behavior remains preserved. The concept is to average the force values among neighboring nodes to mitigate the over- or under-prediction of force. In its absence, especially in scenarios with low density ratios, 'numerical noise' could emerges and intensifies over time, leading to a degradation of results.
- The **HRR collision model** has been developed for LBM, enhancing the stability of the fluid solver through the enforcement of a traceless lattice-Boltzmann tensor. It is the first time at the knowledge of the author that the HRR model is used in FSI. Without it, the cases are restricted to lower Reynolds number.
- **Selective mass scaling** damps the highest non-physical frequencies, primarily originating from numerical instabilities, including noise at the interface due to time lag. This allows an increase in the minimum time step of the structural explicit scheme (as it reduces high frequencies) and, based on observations, stabilizes the coupling. In the present cases, the stable fluid time step has consistently been greater than the stable structure time step. With the use of selective mass scaling, the structure time step can be increased to match the fluid time step, instead of having to reduce the fluid time step and losing computational efficiency.

4. Validation of the weak coupling framework

This section presents a set of validation test cases of increasing complexity (2D rigid, 2D deformable, 3D laminar and turbulent) for the proposed explicit solver, which is shown to remain stable and accurate, whereas these test cases are performed with implicit coupling in literature (except for the sedimenting Particle).

4.1. 2D rigid test-cases

We consider first 2D rigid objects immersed in a fluid flow, experiencing a motion of translation and/or rotation without deformation. For this reason, the finite element solver is replaced by a simple python script. Regarding the stability tools, only the HRR model for the LBM is present, the mass scaling and the smoothing of the forces will be only used by the FEM solver.

4.1.1. Sedimenting particle

A particle, modeled by a disk of diameter D enclosed in a rectangular domain of size $L \times L$, is falling under gravity in a fluid at rest (with lower density than the particle density), as done in [60] and illustrated in (Fig. 6). The set of geometrical and physical parameters is presented in (Table 1).

Table 1
Sedimenting particle parameters.

Parameters	Values
Fluid viscosity (ν)	0.05 m ² /s
Fluid density (ρ_f)	1 kg/m ³
Particle density (ρ_p)	1.5 kg/m ³
Gravity (g)	906.7 m/s ²
Mesh size ($\Delta x_s / \Delta x_f = 1$)	0.005 m
Particle diameter (D)	150 $\cdot \Delta x$
Initial position of the particle center	$[(l + 10\Delta x)/2, 0.125L]$
Spatial discretization of (L)	2850 $\cdot \Delta x$
Spatial discretization of (l)	1470 $\cdot \Delta x$

The particle starts by accelerating until it reaches the equilibrium state between its weight, the buoyancy force and the drag force. The terminal Reynolds number is defined by $Re_t = \frac{U_{max} \cdot D}{\nu}$ with U_{max} the terminal velocity of the particle.

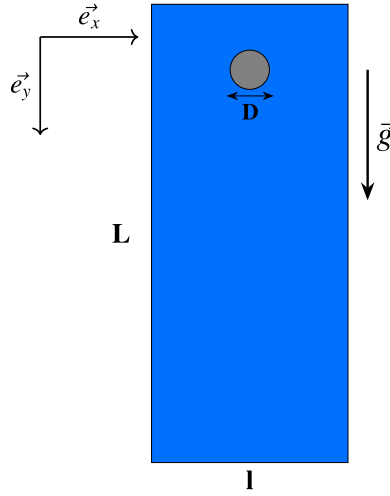


Fig. 6. Geometrical configuration.

We use the following nondimensional numbers, as introduced by [9] that need to be respected:

$$\frac{\rho_p}{\rho_f} = 1.5 \quad G = \frac{Re^2}{Fr} = \frac{D^3 |g|}{\nu^2} = 1.53 \cdot 10^5 \quad (23)$$

The initial position of the particle has a small lateral offset in order to trigger the particle's rotation. A zero velocity boundary condition is imposed on all the walls of the fluid domain. To compute the dynamics of the particle, we use Newton's law, where the sum of the forces applied by the fluid on the particle $\mathbf{F}_{f/p}$ is obtained by IBM, to determine the acceleration $\ddot{\mathbf{u}}_p$ of the particle center. For the angular momentum $\dot{\theta}_p$, the moment $\mathbf{M}_O$ is divided by its inertia $I = \frac{1}{2}m(D/2)^2$, m being the particle mass. We have the following equations for the acceleration:

$$\ddot{\mathbf{u}}_p = \frac{\mathbf{F}_{f/p}}{(\rho_p - \rho_f)V_p} + \mathbf{g} \quad \dot{\theta}_p = \frac{\mathbf{M}_O}{I} \quad (24)$$

The integration of the translational acceleration $\ddot{\mathbf{u}}_p$ is done using an explicit Euler scheme, as follows:

$$\begin{cases} \ddot{u}^{n+1} = \frac{F_p^{n+1}}{(\rho_p - \rho_f)V_p} + g \\ \dot{u}^{n+1} = \dot{u}^{n+1} \cdot \Delta t \\ u^{n+1} = u^{n+1} \cdot \Delta t \end{cases} \quad (25)$$

For the rotation, the moment M_O is defined as the sum of the cross vector for the n nodes composing the particle between the force F_i and the radius vector A_i . The index i represents the current number of the solid node.

$$\mathbf{M}_O = \sum_{i=1}^n \mathbf{F}_i \wedge \mathbf{A}_i \quad (26)$$

Then to calculate and integrate the angular acceleration $\ddot{\theta}^{n+1}$, we also use a first order Euler scheme to determine the new rotation of the particle θ^{n+1} :

$$\begin{cases} \ddot{\theta}^{n+1} = \frac{M_O^{n+1}}{I} \\ \dot{\theta}^{n+1} = \dot{\theta}^{n+1} \cdot \Delta t \\ \theta^{n+1} = \theta^{n+1} \cdot \Delta t \end{cases} \quad (27)$$

The time step for the explicit fluid solver is set to 10^{-5} s.

(Fig. 7) shows the evolution of the particle with the surrounding fluid and (Fig. 8) presents the evolution of the particle velocity as a function of the non-dimensional time. The obtained terminal Reynolds number is 320.1, showing a very good agreement with literature results (Table 2).

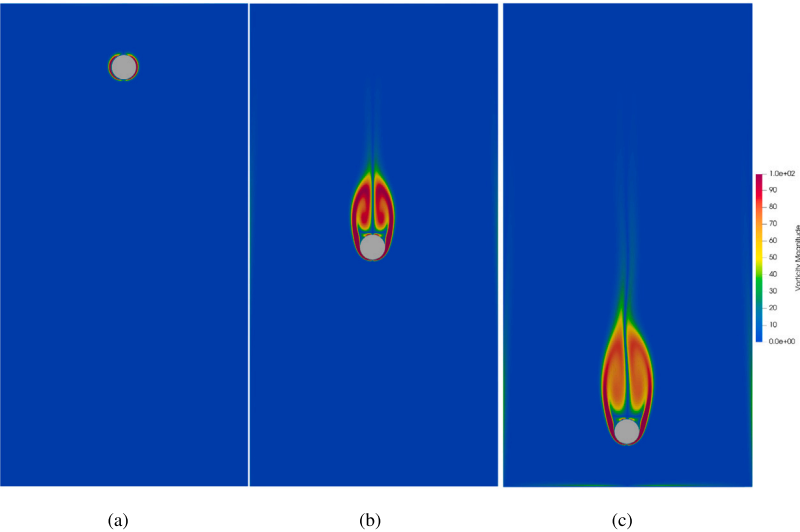


Fig. 7. Vorticity magnitude around the particle at 3 different times. (a) First iteration of the simulation. (b) Acceleration phase of the particle. (c) Stationary state.

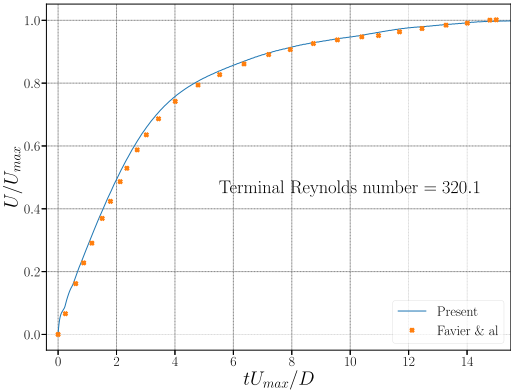


Fig. 8. Evolution of the particle velocity toward stabilization once the force equilibrium is reached.

Table 2 Terminal Reynolds number obtained for the sedimenting particle test case.	
Uhlmann and al	329.8
Favier and al	313.2
Present	320.1

4.1.2. Rigid valve

The second test case focuses on a 2D valve in a channel, as illustrated in (Fig. 9). Compared to the previous test case, a pulsatile flow is imposed at the inlet, the Reynolds number is higher (750) and the density ratio between the fluid and the structure is ≈ 1 which is a challenging for FSI, especially with an explicit coupling algorithm.

A comparison between an experimental and numerical set-up has been performed in [61], and later on by [13] using an implicit coupling scheme.

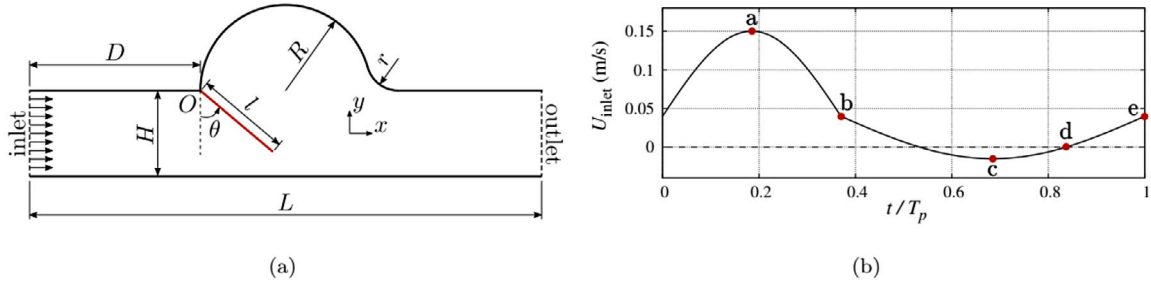


Fig. 9. (a) Geometric configuration. (b) Pulsating inlet velocity.

The motion of the valve is a rotation around the point O along the normal axis of the domain. During the expulsion phase, the valve opens until $\approx 90^\circ$ of inclination. Then a part of the flow goes into the cavity and creates a recirculation pattern occurring at the same time as the expulsion phase. These combined effects tend to close the valve, in phase with the transition to the negative phase of the velocity inlet. The geometrical and physical parameters are defined in the (Table 3).

Table 3

Geometric Parameters		Physical parameters	
Height of the flow channel (H)	0.02 m	Reference velocity (U_{ref})	0.04 m/s
Radius of the transition (r)	0.0075 m	Flow cycle (T_p)	2.45 s
Length of the flow channel (L)	6 H	Inertia (I)	3.9×10^{-6} kg/m ²
Radius of the cavity (R)	H	Fluid viscosity (ν_f)	4×10^{-6} m ² /s
Distance from the inlet (D)	H	Fluid density (ρ_f)	1090 kg/m ³
Valve length (l)	1.07 H	Solid density (ρ_s)	1100 kg/m ³
Valve thickness (e)	0.001 m	Mesh size ($\Delta x_s/\Delta x_f = 1$)	5×10^{-4} m
Initial inclination (θ)	90°	Time step Δt	9.6×10^{-5} s

The pulsating flow at the inlet is given by the function :

$$U_{inlet}(t) = \begin{cases} U_{ref} + U_{amp} \sin\left(\frac{2\pi t/T_p}{0.74}\right), & \text{if } 0 < t/T_p \leq 0.37, \\ U_{ref} + \frac{U_{amp}}{2} \sin\left(\frac{2\pi(t/T_p + 0.26)}{1.26}\right), & \text{if } 0.37 < t/T_p \leq 1, \end{cases} \quad (28)$$

The amplitude of the inlet velocity U_{amp} is set to 0.11 m/s, and the period to $T_p = 2.45$ s. The peak value of velocity at the inlet is 0.15 m/s which corresponds to a Reynolds number of $Re = U_{max}l/\nu_f \approx 750$. For the outlet, we fix the fluid density at its initial value. The inertial moment of the rod is set to $I = \rho_s l^3 e/3$.

The comparison between the measured velocity fields by Particle Image Velocimetry (PIV) by [61] and our numerical results show a good agreement on the flow pattern and the rotation angle in (Fig. 10). The periodicity of the results and the comparison of the angle evolution with the literature is presented in (Fig. 11). We report here a very good stability of the simulation over several periods without any sign of accumulation of spurious numerical noise, even is the density ratio is ≈ 1 . It is worth noting also that the time evolution of the angle is excellent with the one obtained in [13], where a strong coupling implicit algorithm was mandatory to avoid divergence.

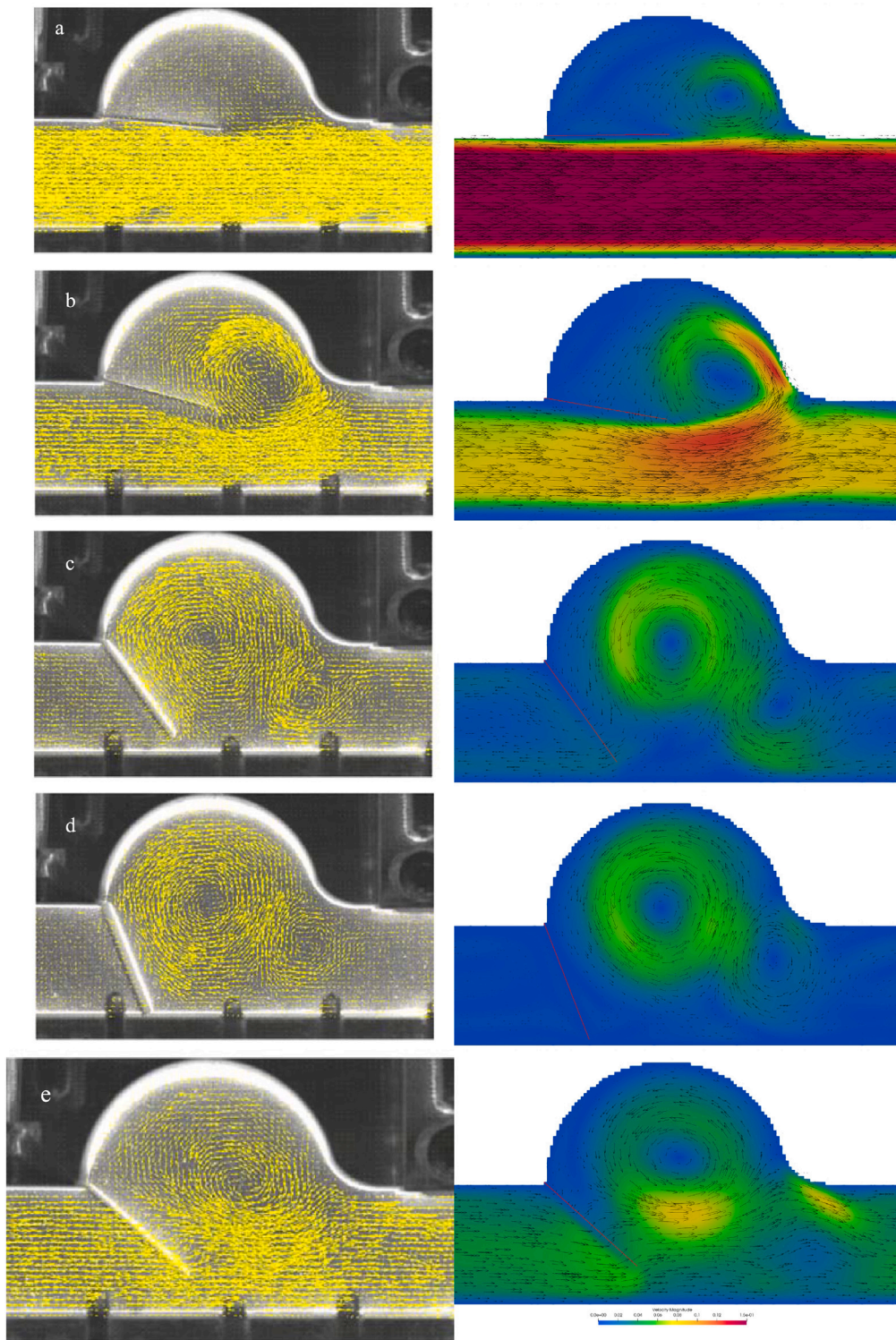


Fig. 10. Comparison between the velocity field obtained in [61] by PIV and the present numerical results. The instant a,b,c,d,e correspond to specific moments of the inlet (Fig. 9).

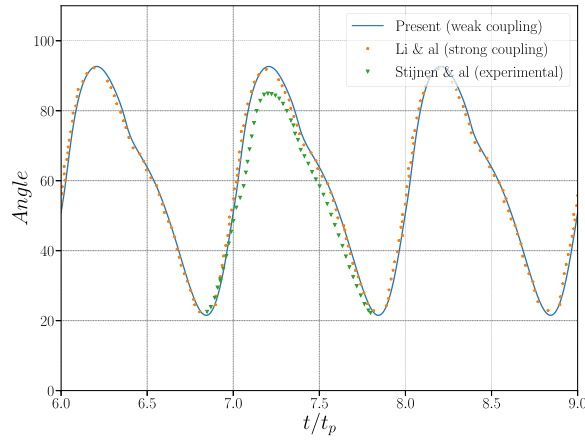


Fig. 11. Time evolution of the angle θ in the 2D rigid valve test-case.

4.2. 2D deformable test-case

We consider here a 2D elastic beam immersed in a parabolic flow and we use from now the FEM solver to solve the structure dynamics. This test case is a reference for FSI validation algorithm [62]. It is composed by a fixed cylinder with an elastic beam attached behind it, in a parabolic flow (Fig. 12) and (Table 4). The flow around the cylinder leads to vortex shedding in the wake of the cylinder, to which the beam reacts because of the periodic pressure variation and tends to oscillate. At the stationary state, the movement of the beam is periodic and in resonance with the flow. The displacement of the tip (point A in (Fig. 12)) is compared to the reference data.

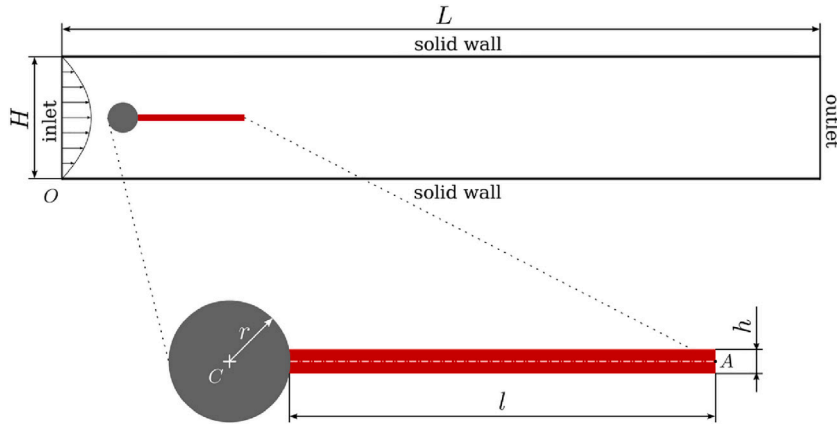


Fig. 12. Setup of the computational domain from [16].

Table 4

Geometric parameters of the 2D deformable test case.

Parameters	Values
Length of the flow channel (L)	2.5 m
Height of the flow channel (H)	0.41 m
Radius of the cylinder (r)	0.05 m
Length of the solid bar (l)	0.35 m
Height of the solid bar (h)	0.02 m
Coordinate of the point (O)	(0.0 m, 0.0 m)
Coordinate of the cylinder center (C)	(0.2 m, 0.2 m)
Coordinate of the point (A)	(0.6 m, 0.2 m)

We consider the FSI2 and FSI3 configurations detailed in (Table 5). They have the same geometry but different Reynolds number and material properties, the FSI3 being the most challenging case, because of its higher Reynolds number and density ratio of 1.

In the framework of LBM, we make sure that the Mach number for both cases is under 0.3 to reduce compressibility errors, by adjusting the value of the sound speed c_s in the fluid in Eq. (5).

Table 5

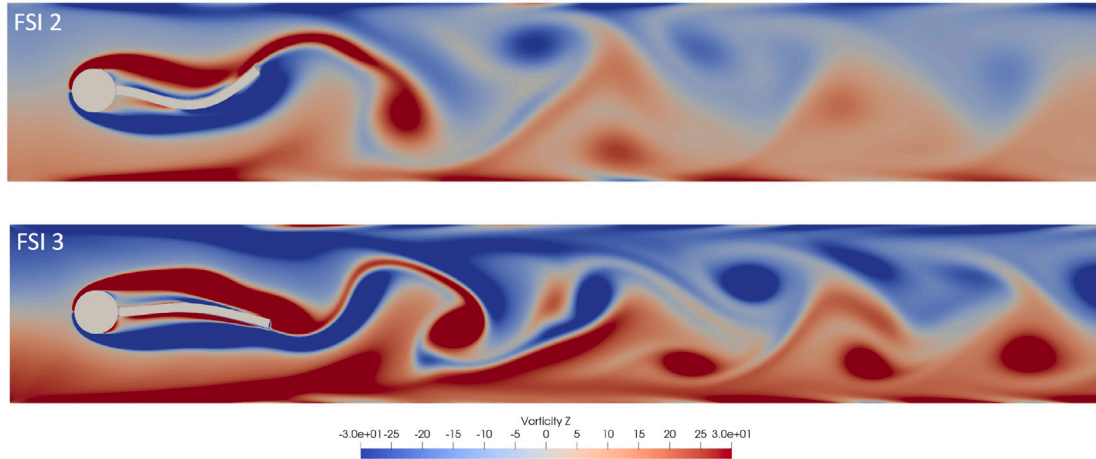
Material and numerical parameters of the 2D deformable test case.

	FSI2 (R e=100)	FSI3 (R e=200)
Fluid density (ρ_f)	10^3 kg/m^3	10^3 kg/m^3
Fluid kinematic viscosity (ν_f)	$10^{-3} \text{ m}^2/\text{s}$	$10^{-3} \text{ m}^2/\text{s}$
Solid density (ρ_s)	10^4 kg/m^3	10^3 kg/m^3
Solid Poisson ratio (ν_s)	0.4	0.4
Solid Young's modulus (E_s)	$1.4 \times 10^6 \text{ Pa}$	$5.6 \times 10^6 \text{ Pa}$
Solid-to-fluid density ratio (ρ_s/ρ_f)	10	1
Fluid velocity ($\bar{U}$)	1 m/s	2 m/s
Mach number (Ma)	0.20	0.22
Mesh size ($\Delta x_s/\Delta x_f = 1$)	0.002 m	0.001 m
Time step Δt	$2.3 \times 10^{-4} \text{ s}$	$6.4 \times 10^{-5} \text{ s}$
Solid time step factor	18	60
Number of nodes for the force smoothing	5	5

For the outlet boundary condition, we fix the fluid density to its initial value. For the inlet velocity, a parabolic profile is imposed with a temporal initial ramp during 2s to smooth the transient phase:

$$\begin{cases} v_f^x(0, y, t) = 1.5\bar{U} \frac{y(H-y)}{(0.5H)^2} \frac{1-\cos(0.5\pi t)}{2} & \text{if } t < 2.0 \text{ s} \\ v_f^x(0, y, t) = 1.5\bar{U} \frac{y(H-y)}{(0.5H)^2} & \text{otherwise} \end{cases} \quad (29)$$

(Fig. 13) shows the results obtained for FSI2 and FSI3 configurations, characterized by a periodic vortex shedding in the wake of the object, with higher vorticity and larger vortex frequency for FSI3 due to its higher Reynolds number. Once the stationary state is reached, we compare the displacement of the tip A along the x and y axis to the reference (Fig. 14), the frequencies are provided in (Table 6).

**Fig. 13.** Vorticity of the flow for the FSI2 and FSI3 configurations.**Table 6**

Oscillations frequencies compared to a monolithic solver.

FSI2	$u_x \text{ Freq. (Hz)}$	$u_y \text{ Freq. (Hz)}$
Present	3.8	1.9
Ref [62]	3.8	2
FSI3	$u_x \text{ Freq. (Hz)}$	$u_y \text{ Freq. (Hz)}$
Present	11.4	5.6
Ref [62]	10.9	5.3

A good agreement is found in amplitude and frequency demonstrating the accuracy of the method. We can notice a discrepancy, specific to the FSI3 case, concerns the non-symmetrical displacement amplitude cycle along the x axis contrary to the reference which is perfectly symmetrical. This behavior has already been observed on other FSI3 comparisons such as [63]. Given that the u_x displacement is very sensitive and has a negligible influence on the flow compared to the u_y displacement, it does not affect the global results and we do not consider this difference as being restrictive.

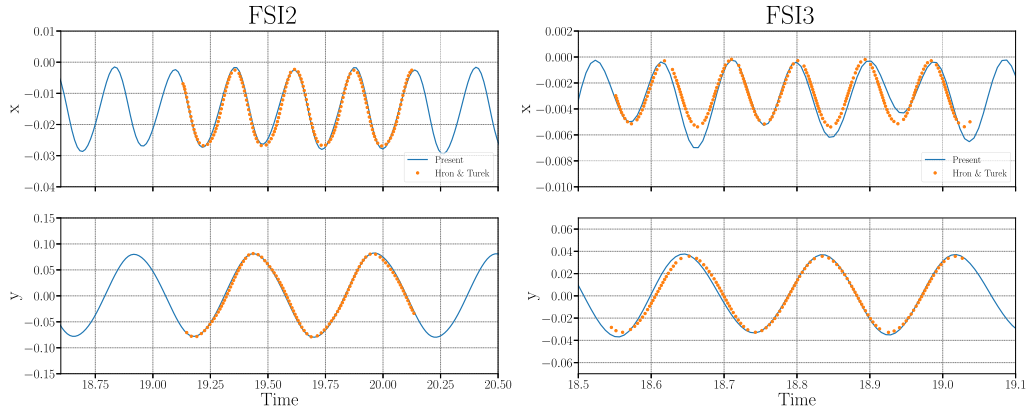


Fig. 14. Displacement of the point A for FSI2 and FSI3 cases.

Thus, we report here a good agreement on FSI2 and FSI3 configurations compared to the reference case, using a purely explicit coupling even in the case of density ratio equal to 1.

4.3. 3D test-cases

We compare here our results with literature on two large-scale 3D cases, to assess the capacity of the present framework to handle complex configurations with a reasonable computational time.

4.3.1. Flexible plate in cross flow

This test case is commonly used to study experimentally and theoretically plant posture, which plays a key role in the health of aquatic vegetation, by setting drag, controlling light availability, and mediating the exchange of nutrients and oxygen [64]. It is identified in literature [65] that an implicit coupling algorithm is necessary to reproduce accurately the FSI dynamics of this test case. It consists in a plate of dimensions $[h, b, L]$ immersed in a rectangular fluid domain extending from $[-5b, -8b, -8.5b]$ to $[16b, 8b, 8.5b]$, as illustrated in (Fig. 15). The center of the plate is located initially at the origin, one side of the plate is clamp-mounted and the other one is free. The flow is uniform with a velocity U_0 in the cross flow direction. For the outlet we set the fluid density constant and equal to the initial fluid density ρ_f . We use the set of parameters described in [65] and listed in (Table 7).

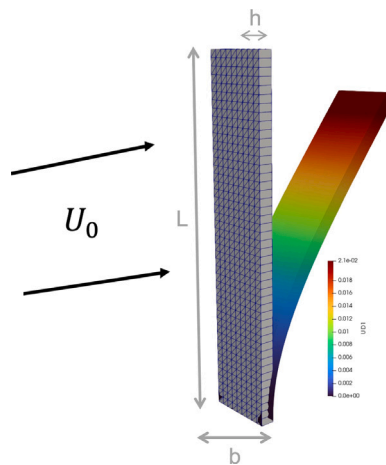


Fig. 15. Visualization of the flexible plate of size $[h, b, L]$ in a uniform flow of velocity U_0 , the colored plate represents the final state with the displacement along the x axis.

In the FEM model, we use 2D shell elements extruded into 3D by the structure solver, and the thickness of the plate is thus composed of only 1 element, as illustrated in (Fig. 15). This approximation can be done because of the size ratio of the object (Table 7).

Table 7

Physic parameters		Numerical parameters	
Reynolds ($U_0 b/\nu$)	1600	Mach number (Ma)	0.17
Size ratio (L/b)	5	Mesh size ($\Delta x_s/\Delta x_f = 1$)	0.00105 m
Size ratio (h/b)	0.2	Time step (Δt)	6.1×10^{-5} s
Young modulus ($E/\rho_f U_0^2$)	19 054.9	Simulation time	4 s
Density ratio (ρ_s/ρ_f)	0.678	Solid time step factor	98
Poisson coefficient (ν_s)	0.4	Number of nodes for the force smoothing	5

After a transient oscillatory regime, the balance between the force applied by the fluid and the internal reaction force of the solid is reached, leading to a quasi-steady state for the beam, represented on (Fig. 16). The evolution of the drag coefficient $C_d = F_x/0.5\rho_f U_0^2 bL$ and the deflection D_x/b and D_z/b of the plate along x and z axis is therefore almost constant and we compare them to the literature (Table 8).

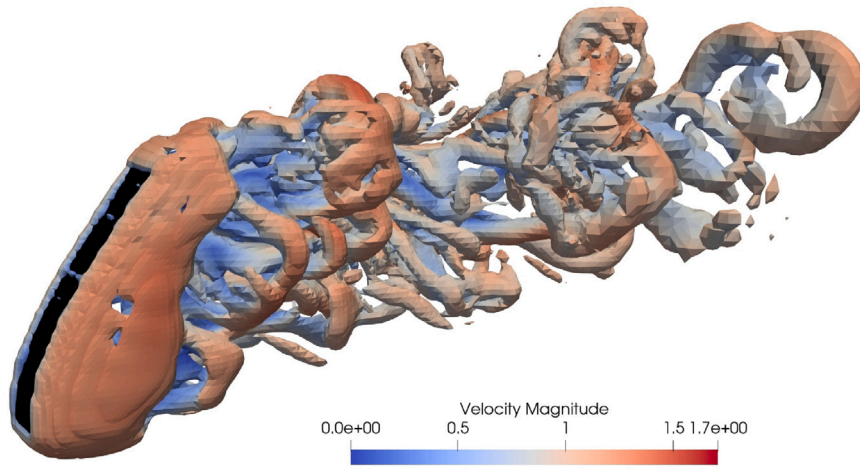


Fig. 16. Visualization of the Q-criterion in the wake of the black plate colored by the velocity magnitude. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 8

Comparison with experimental and numerical data from the literature of the drag coefficient and deflection along x and z axis. The 10% range of error on the drag of Luhar and Nepf are coming from the measure.

Case	C_d	D_x/b	D_z/b
Luhar and Nepf [64] (exp)	1.15 (10%)	2.14	0.59
Tian and al [65] (num)	1.03	2.12	0.54
Present	1.05	2.1	0.53

A very good agreement is observed compared to the experimental and numerical data, which shows the ability of our purely explicit FSI solver to handle this 3D reference test case.

4.3.2. 3D elastic beam immersed in a parabolic turbulent flow

We examine here a FSI test case with complex flow at higher Reynolds number and on a large computational domain (1 400 000 fluid nodes and 16 000 solid nodes). The same parameters as the FS13 case are used, excepting the Reynolds number, which is set to 10 000, an extrusion of the object and fluid domain of 0.048 m to perform 3D computations, mandatory to consider turbulent flows. Finally, the lateral channel walls are assumed to be slip walls.

We perform an initialization phase, with fixed geometry to let the turbulent flow settle, before releasing the beam. To account for the unresolved scales of motion in the turbulent case, a subgrid model is used which is a variant of the Smagorinsky model called the “shear-improved Smagorinsky” proposed in [66]. In this model, the calculation of the turbulent viscosity is:

$$\nu_t = (C_s^2 \Delta x)^2 (|S| - S) \quad (30)$$

with C_S the Smagorinsky constant, Δx the local grid spacing and $|S|$ the norm of the rate of strain tensor. The correcting term S is the norm of the low-pass filtering (in time) of the rate of strain tensor. This filtering is explained in detail in [66] and is independent of the test case. In the LBM, the turbulent viscosity is taken into account in the calculation of the physical relaxation time τ^S :

$$\tau^S = \frac{\nu + \nu_t}{c_s^2} + \frac{1}{2} \Delta t \quad (31)$$

with ν and ν_t the molecular viscosity and turbulent viscosity respectively.

We can observe on (Fig. 17) that the Q-criterion reveals a turbulent wake around the cylinder with some detachment behind the beam. The magnitude of the displacement (Fig. 18) is reduced compared to the FSI3 2D laminar case (see Fig. 14), this is also observed in [30].

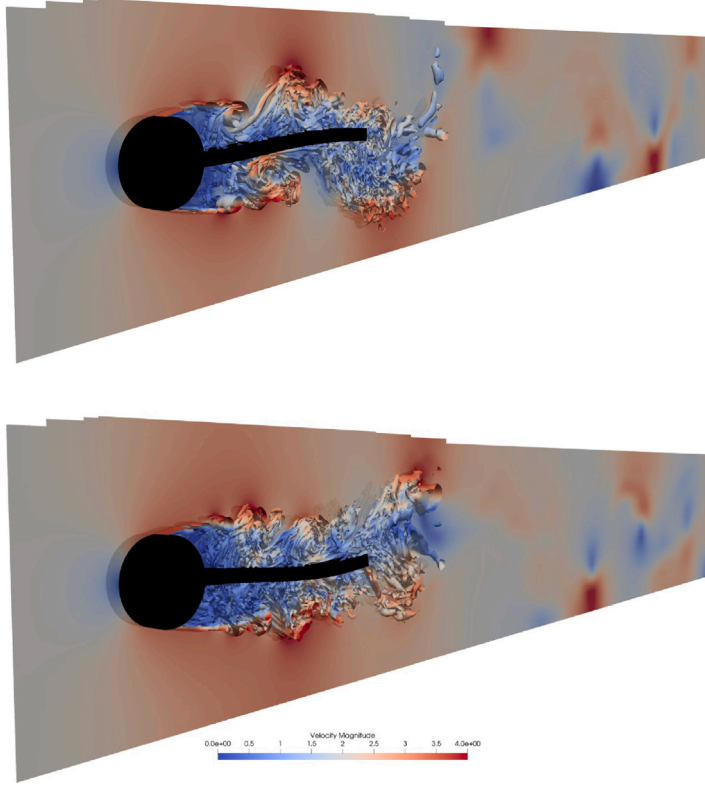


Fig. 17. 3D cylinder+beam immersed in a turbulent flow with visualization of the Q-criterion and of a slice of the fluid domain colored by velocity.

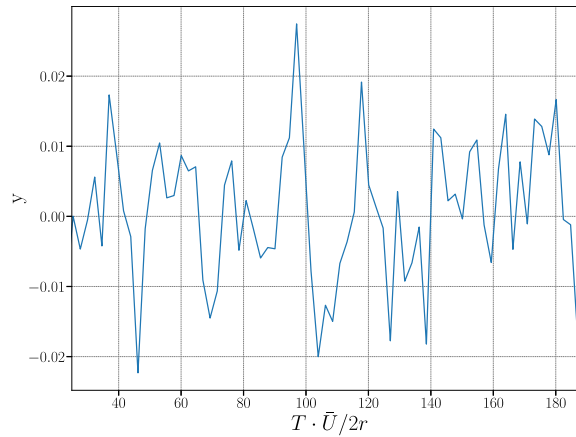


Fig. 18. Vertical displacement of the point A at the tip of the 3D beam with a dimensionless time.

The turbulent flow does not lead to a perfectly periodic vortex detachment, which normally corresponds to the maximum amplitude of the beam bending. Here, the turbulent flow mixes the energy between the different scales of motion of the fluid and prevents the smooth vortex shedding required to deform significantly and periodically the beam.

We compare our results on the aerodynamic coefficients and Strouhal number with the numerical results of [30] obtained using a semi-implicit coupling based on the pressure term in the Navier–Stokes equation.

A good agreement with the literature data is observed for the amplitude of the lift in (Fig. 19(a)). An exact superposition with the reference is not expected because of the unsteadiness of the turbulent flow. We obtain a mean drag of 1.859 compared to 1.215 in [30]. This over-prediction can be due to the IBM with direct forcing and has already been reported in [67] but it does not impact the dynamics of the flow.

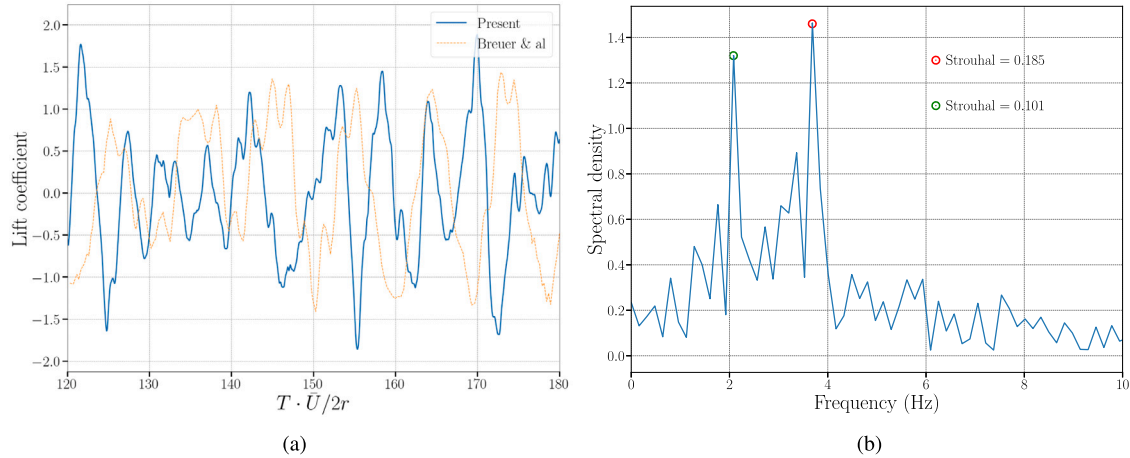


Fig. 19. (a) Lift coefficient of the turbulent case with dimensionless time. (b) Vortex shedding frequency analysis.

Consistent results are also found in the frequency domain, the Fourier analysis provided in (Fig. 19(b)) shows two dominant modes. The first one corresponds to the periodic detachment of the boundary layer around the cylinder and has a Strouhal of 0.185 with the present solver consistent with the value of 0.175 in [30]. The second mode corresponding to the periodic oscillation of the beam has a Strouhal of 0.101 with the present solver compared to 0.103 in [30].

Finally, the LBM-IBM-FEM framework proposed here with an explicit coupling shows excellent stability properties and good agreement compared to the literature on a challenging 3D turbulent FSI configuration with a density ratio of 1.

5. Conclusion

In this work, we propose a FSI solver based on a Lattice-Boltzmann, Immersed Boundary and Finite Element methods coupled with a stabilized explicit algorithm (CSS). This combination of methods is effective and relatively easy to implement. While in literature the use of the CSS algorithm is restricted to weak coupling physical configurations with $(\rho_s/\rho_f) \neq 1$, the proposed solver allows for stable and accurate results on challenging configurations including 3D turbulent flow with a density ratio equal to 1. It introduces a new process of stabilization based on three numerical treatments which prevent the accumulation of spurious numerical errors at the fluid–solid interface:

- 1. the Laplacian smoothing which is an average of the force values between the neighboring nodes,
- 2. the HRR collision model which enforces a traceless lattice-Boltzmann tensor and increases the stability of the fluid solver,
- 3. a selective mass scaling in the FEM to damp the highest non-physical frequencies.

This work provides an insightful and novel FSI algorithm to tackle strongly coupled problems involving complex 3D turbulent conditions. We observed a significant improvement in results and stability by using the three features previously mentioned (Laplacian smoothing, HRR model and mass scaling). Future developments will apply this FSI solver to realistic configurations in aeronautics and biological flows with the simulation of the aortic valve.

CRediT authorship contribution statement

Tom Fringand: Writing – review & editing, Writing – original draft, Validation, Software, Investigation. **Isabelle Cheylan:** Writing – review & editing, Validation, Software, Methodology, Investigation, Conceptualization. **Marien Lenoir:** Writing – review & editing, Visualization, Conceptualization. **Loic Mace:** Writing – review & editing, Visualization, Supervision, Funding acquisition, Conceptualization. **Julien Favier:** Writing – review & editing, Visualization, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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References

- [1] J.R. Meneghini, F. Saltara, R. de Andrade Fregonesi, C.T. Yamamoto, E. Casaprima, J.A. Ferrari, Numerical simulations of VIV on long flexible cylinders immersed in complex flow fields, *Eur. J. Mech. B/Fluids* 23 (1) (2004) 51–63, <http://dx.doi.org/10.1016/j.euromechflu.2003.09.006>, URL <https://www.sciencedirect.com/science/article/pii/S0997754603000888> Bluff Body Wakes and Vortex-Induced Vibrations.
- [2] R. Kamakoti, W. Shyy, Fluid-structure interaction for aeroelastic applications, *Prog. Aerosp. Sci.* 40 (8) (2004) 535–558, <http://dx.doi.org/10.1016/j.paerosci.2005.01.001>, URL <https://www.sciencedirect.com/science/article/pii/S0376042105000084>.
- [3] B. Becsek, L. Pietrasanta, D. Obrist, Turbulent systolic flow downstream of a bioprosthetic aortic valve: Velocity spectra, wall shear stresses, and turbulent dissipation rates, *Front. Physiol.* (2020) 577188, <http://dx.doi.org/10.3389/fphys.2020.577188>.
- [4] M.-C. Hsu, Y. Bazilevs, Fluid-structure interaction modeling of wind turbines: simulating the full machine, *Comput. Mech.* (2012) <http://dx.doi.org/10.1007/s00466-012-0772-0>.
- [5] M. Souli, A. Ouahsine, L. Lewin, ALE formulation for fluid-structure interaction problems, *Comput. Methods Appl. Mech. Engrg.* 190 (5) (2000) 659–675, [http://dx.doi.org/10.1016/S0045-7825\(99\)00432-6](http://dx.doi.org/10.1016/S0045-7825(99)00432-6), URL <https://www.sciencedirect.com/science/article/pii/S0045782599004326>.
- [6] D. Boffi, L. Gastaldi, Stability and geometric conservation laws for ALE formulation, *Comput. Methods Appl. Mech. Engrg.* 193 (2004) 4717–4739, <http://dx.doi.org/10.1016/j.cma.2004.02.020>.
- [7] T.J. Hughes, W.K. Liu, T.K. Zimmermann, Lagrangian-Eulerian finite element formulation for incompressible viscous flows, *Comput. Methods Appl. Mech. Engrg.* 29 (3) (1981) 329–349, [http://dx.doi.org/10.1016/0045-7825\(81\)90049-9](http://dx.doi.org/10.1016/0045-7825(81)90049-9), URL <https://www.sciencedirect.com/science/article/pii/0045782581900499>.
- [8] C. Peskin, Flow patterns around heart valves: A numerical method, *J. Comput. Phys.* 10 (2) (1972) 252–271, [http://dx.doi.org/10.1016/0021-9991\(72\)90065-4](http://dx.doi.org/10.1016/0021-9991(72)90065-4).
- [9] J. Favier, A. Revell, A. Pinelli, A Lattice Boltzmann-Immersed Boundary method to simulate the fluid interaction with moving and slender flexible objects", *J. Comput. Phys.* 261 (2014) 145–161, <http://dx.doi.org/10.1016/j.jcp.2013.12.052>.
- [10] L. Wang, Locomotion of a self-propulsive pitching plate in a quiescent viscous fluid, *Proc. Inst. Mech. Eng. C* 235 (2) (2021) 342–350, <http://dx.doi.org/10.1177/0954406220903338>.
- [11] Q. Huang, Low Reynolds Number Turbulent FSI and Its Applications in Biological Flows (Ph.D. thesis), 2021, <http://dx.doi.org/10.26190/unsworks/3993>.
- [12] L. Wang, Z. Liu, M. Rajamuni, Recent progress of lattice Boltzmann method and its applications in fluid-structure interaction, *Proc. Inst. Mech. Eng. C* (2022) 095440622210775, <http://dx.doi.org/10.1177/09544062221077583>.
- [13] Z. Li, G. Oger, D. Le Touzé, A partitioned framework for coupling LBM and FEM through an implicit IBM allowing non-conforming time-steps: Application to fluid-structure interaction in biomechanics, *J. Comput. Phys.* 449 (C) (2022) <http://dx.doi.org/10.1016/j.jcp.2021.110786>.
- [14] R.-N. Hua, L. Zhu, X.-Y. Lu, Locomotion of a flapping flexible plate, *Phys. Fluids* 25 (12) (2013) 121901, <http://dx.doi.org/10.1063/1.4832857>, [arXiv:https://pubs.aip.org/aip/pof/article-pdf/doi/10.1063/1.4832857/15943697/121901_1_online.pdf](https://pubs.aip.org/aip/pof/article-pdf/doi/10.1063/1.4832857/15943697/121901_1_online.pdf).
- [15] R.-N. Hua, L. Zhu, X.-Y. Lu, Dynamics of fluid flow over a circular flexible plate, *J. Fluid Mech.* 759 (2014) 56–72, <http://dx.doi.org/10.1017/jfm.2014.571>.
- [16] Z. Li, J. Favier, A non-staggered coupling of finite element and lattice Boltzmann methods via an immersed boundary scheme for fluid-structure interaction, *Comput. & Fluids* 143 (2017) 90–102, <http://dx.doi.org/10.1016/j.compfluid.2016.11.008>, URL <https://www.sciencedirect.com/science/article/pii/S0045793016303589>.
- [17] Y. Cheng, H. Zhang, C. Liu, Immersed boundary-Lattice Boltzmann coupling scheme for fluid-structure interaction with flexible boundary, *Commun. Comput. Phys.* 9 (5) (2011) 1375–1396, <http://dx.doi.org/10.4208/cicp.301009.211210s>.
- [18] Q. Huang, Z. Liu, L. Wang, S. Ravi, J. Young, J.C.S. Lai, F.-B. Tian, Streamline penetration, velocity error, and consequences of the feedback immersed boundary method, *Phys. Fluids* 34 (9) (2022) 097101, <http://dx.doi.org/10.1063/5.0101584>, [arXiv:https://pubs.aip.org/aip/pof/article-pdf/doi/10.1063/5.0101584/16587991/097101_1_online.pdf](https://pubs.aip.org/aip/pof/article-pdf/doi/10.1063/5.0101584/16587991/097101_1_online.pdf).
- [19] W.-X. Huang, F.-B. Tian, Recent trends and progress in the immersed boundary method, *Proc. Inst. Mech. Eng. C* 233 (23–24) (2019) 7617–7636, <http://dx.doi.org/10.1177/0954406219842606>.
- [20] K. Wong, P. Thavornpattana, S. Cheung, J. Tu, Numerical stability of partitioned approach in fluid-structure interaction for a deformable thin-walled vessel, *Comput. Math. Methods Med.* 2013 (2013) 638519, <http://dx.doi.org/10.1155/2013/638519>.
- [21] C. Förster, W.A. Wall, E. Ramm, Artificial added mass instabilities in sequential staggered coupling of nonlinear structures and incompressible viscous flows, *Comput. Methods Appl. Mech. Engrg.* 196 (7) (2007) 1278–1293, <http://dx.doi.org/10.1016/j.cma.2006.09.002>, URL <https://www.sciencedirect.com/science/article/pii/S0045782506002544>.
- [22] Z. Li, G. Oger, D. Le Touzé, A partitioned framework for coupling LBM and FEM through an implicit IBM allowing non-conforming time-steps: Application to fluid-structure interaction in biomechanics, *J. Comput. Phys.* 449 (2022) 110786, <http://dx.doi.org/10.1016/j.jcp.2021.110786>, URL <https://www.sciencedirect.com/science/article/pii/S0021999121006811>.
- [23] J.-F. Gerbeau, M. Vidrascu, A quasi-Newton algorithm based on a reduced model for fluid-structure interaction problems in blood flows, *ESAIM Math. Model. Numer. Anal.* 37 (4) (2003) 631–647, <http://dx.doi.org/10.1051/m2an:2003049>, URL <https://hal.inria.fr/hal-00694625>.
- [24] E. Burman, M.A. Fernández, Stabilization of explicit coupling in fluid-structure interaction involving fluid incompressibility, *Comput. Methods Appl. Mech. Eng.* 198 (5–8) (2009) 766–784, <http://dx.doi.org/10.1016/j.cma.2008.10.012>, URL <https://hal.inria.fr/inria-00247409>.
- [25] J. Degroote, K.-J. Bathe, J. Vierendeels, Performance of a new partitioned procedure versus a monolithic procedure in fluid-structure interaction, *Comput. Struct.* 87 (11) (2009) 793–801, <http://dx.doi.org/10.1016/j.compstruc.2008.11.013>, URL <https://www.sciencedirect.com/science/article/pii/S0045794908002605> Fifth MIT Conference on Computational Fluid and Solid Mechanics.

- [26] M. Heil, An efficient solver for the fully coupled solution of large-displacement fluid–structure interaction problems, *Comput. Methods Appl. Mech. Engrg.* 193 (1) (2004) 1–23, <http://dx.doi.org/10.1016/j.cma.2003.09.006>, URL <https://www.sciencedirect.com/science/article/pii/S0045782503005164>.
- [27] S. Rugonyi, K. Bathe, On finite element analysis of fluid flows fully coupled with structural interactions, *CMES Comput. Model. Eng. Sci.* 2 (2) (2001) 195–212.
- [28] T.E. Tezduyar, Finite element methods for fluid dynamics with moving boundaries and interfaces, in: *Encyclopedia of Computational Mechanics*, John Wiley & Sons, Ltd, 2004, <http://dx.doi.org/10.1002/0470091355.ecm069>, arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1002/0470091355.ecm069> URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/0470091355.ecm069>.
- [29] K.-J. Bathe, H. Zhang, Finite element developments for general fluid flows with structural interactions, *Internat. J. Numer. Methods Engrg.* 60 (1) (2004) 213–232, <http://dx.doi.org/10.1002/nme.959>, arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1002/nme.959> URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/nme.959>.
- [30] M. Breuer, G. De Nayer, M. Münch, T. Gallinger, R. Wüchner, Fluid–structure interaction using a partitioned semi-implicit predictor–corrector coupling scheme for the application of large-eddy simulation, *J. Fluids Struct.* 29 (2012) 107–130, <http://dx.doi.org/10.1016/j.jfluidstructs.2011.09.003>, URL <https://www.sciencedirect.com/science/article/pii/S0889974612000096>.
- [31] B. Connell, D. Yue, Flapping dynamics of a flag in a uniform stream, *J. Fluid Mech.* 581 (2007) 33–67, <http://dx.doi.org/10.1017/S0022112007005307>.
- [32] L. Xu, F.-B. Tian, J. Young, J.C. Lai, A novel geometry-adaptive Cartesian grid based immersed boundary–lattice Boltzmann method for fluid–structure interactions at moderate and high Reynolds numbers, *J. Comput. Phys.* 375 (2018) 22–56, <http://dx.doi.org/10.1016/j.jcp.2018.08.024>, URL <https://www.sciencedirect.com/science/article/pii/S0021999118305473>.
- [33] J. Jacob, O. Malaspinas, P. Sagaut, A new hybrid recursive regularised Bhatnagar–Gross–Krook collision model for Lattice Boltzmann method-based large eddy simulation, *J. Turbul.* 19 (11–12) (2018) 1051–1076, <http://dx.doi.org/10.1080/14685248.2018.1540879>.
- [34] L. Olovsson, K. Simonsson, M. Unosson, Selective mass scaling for explicit finite element analyses, *Int. J. Numer. Methods Eng.* 63 (2005) 1436–1445, <http://dx.doi.org/10.1002/nme.1293>.
- [35] T. Krüger, H. Kusumaatmaja, A. Kuzmin, O. Shardt, G. Silva, E. Viggen, *The Lattice Boltzmann Method. Principles and Practice*, Springer, 2017.
- [36] J. Jacob, P. Sagaut, Wind comfort assessment by means of large eddy simulation with lattice Boltzmann method in full scale city area, *Build. Environ.* 139 (2018) 110–124, <http://dx.doi.org/10.1016/j.buildenv.2018.05.015>.
- [37] L. Merlier, J. Jacob, P. Sagaut, Lattice-Boltzmann Large-Eddy simulation of pollutant dispersion in street canyons including tree planting effects, *Atmos. Environ.* 195 (2018) <http://dx.doi.org/10.1016/j.atmosenv.2018.09.040>.
- [38] Y. Feng, P. Boivin, J. Jacob, P. Sagaut, Hybrid recursive regularized lattice Boltzmann simulation of humid air with application to meteorological flows, *Phys. Rev. E* 100 (2) (2019) 023304, <http://dx.doi.org/10.1103/PhysRevE.100.023304>.
- [39] Y. Feng, P. Boivin, J. Jacob, P. Sagaut, Hybrid recursive regularized thermal lattice Boltzmann model for high subsonic compressible flows, *J. Comput. Phys.* 394 (2019) 82–99, <http://dx.doi.org/10.1016/j.jcp.2019.05.031>.
- [40] S. Guo, Y. Feng, J. Jacob, P. Sagaut, An efficient lattice Boltzmann method for compressible aerodynamics on D3Q19 lattice, *J. Comput. Phys.* 418 (2020) 109570, <http://dx.doi.org/10.1016/j.jcp.2020.109570>.
- [41] Z. Guo, C. Zheng, B. Shi, Discrete lattice effects on the forcing term in the lattice Boltzmann method, *Phys. Rev. E* 65 (2002) 046308, <http://dx.doi.org/10.1103/PhysRevE.65.046308>.
- [42] S. Gsell, U. D’Ortona, J. Favier, Explicit and viscosity-independent immersed-boundary scheme for the lattice Boltzmann method, *Phys. Rev. E* 100 (3) (2019) <http://dx.doi.org/10.1103/PhysRevE.100.033306>.
- [43] Z. Li, J. Favier, U. D’Ortona, S. Poncet, An immersed boundary-lattice boltzmann method for single and multi-component fluid flows, *J. Comput. Phys.* 304 (2016) 422–440, <http://dx.doi.org/10.1016/j.jcp.2015.10.026>.
- [44] I. Cheylan, J. Favier, P. Sagaut, Immersed boundary conditions for moving objects in turbulent flows with the lattice-Boltzmann method, *Phys. Fluids* 33 (2021) 095101, <http://dx.doi.org/10.1063/5.0062575>.
- [45] C.S. Peskin, The immersed boundary method, *Acta Numer.* 11 (2002) 479–517, <http://dx.doi.org/10.1017/S0962492902000077>.
- [46] X. Yang, X. Zhang, Z. Li, G.-W. He, A smoothing technique for discrete delta functions with application to immersed boundary method in moving boundary simulations, *J. Comput. Phys.* 228 (20) (2009) 7821–7836, <http://dx.doi.org/10.1016/j.jcp.2009.07.023>, URL <https://www.sciencedirect.com/science/article/pii/S0021999109004136>.
- [47] A. Roma, C. Peskin, M. Berger, An adaptive version of the immersed boundary method, *J. Comput. Phys.* 153 (2) (1999) 509–534, <http://dx.doi.org/10.1006/jcph.1999.6293>.
- [48] Y. Peng, L.-S. Luo, A comparative study of immersed-boundary and interpolated bounce-back methods in LBE, *Prog. Comput. Fluid Dyn.* 8 (2008) 156–167, <http://dx.doi.org/10.1504/PCFD.2008.018086>.
- [49] I. Cheylan, T. Fringand, J. Jacob, J. Favier, Analysis of the immersed boundary method for turbulent fluid–structure interaction with Lattice Boltzmann method, *J. Comput. Phys.* 492 (2023) 112418, <http://dx.doi.org/10.1016/j.jcp.2023.112418>, URL <https://www.sciencedirect.com/science/article/pii/S0021999123005132>.
- [50] T. Kempe, J. Fröhlich, An improved immersed boundary method with direct forcing for the simulation of particle laden flows, *J. Comput. Phys.* 231 (9) (2012) 3663–3684, <http://dx.doi.org/10.1016/j.jcp.2012.01.021>.
- [51] T. Seta, R. Rojas, K. Hayashi, A. Tomiyama, Implicit-correction-based immersed boundary–lattice Boltzmann method with two relaxation times, *Phys. Rev. E* 89 (2014) 023307, <http://dx.doi.org/10.1103/PhysRevE.89.023307>.
- [52] A. Pinelli, I. Naqavi, U. Piomelli, J. Favier, Immersed-boundary methods for general finite-difference and finite-volume Navier–Stokes solvers, *J. Comput. Phys.* 229 (24) (2010) 9073–9091, <http://dx.doi.org/10.1016/j.jcp.2010.08.021>.
- [53] G. Dhondt, *The Finite Element Method for Three-Dimensional Thermomechanical Applications*, John Wiley & Sons, 2004, <http://dx.doi.org/10.1002/0470021217>.
- [54] I. Miranda, R.M. Ferencz, T.J.R. Hughes, An improved implicit-explicit time integration method for structural dynamics, *Earthq. Eng. Struct. Dyn.* 18 (5) (1989) 643–653, <http://dx.doi.org/10.1002/eqe.4290180505>, arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1002/eqe.4290180505> URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/eqe.4290180505>.
- [55] L. Olovsson, K. Simonsson, Iterative solution technique in selective mass scaling, *Commun. Numer. Methods. Eng.* 22 (1) (2006) 77–82, <http://dx.doi.org/10.1002/cnm.806>, arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1002/cnm.806> URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/cnm.806>.
- [56] C. Farhat, K.G. van der Zee, P. Geuzaine, Provably second-order time-accurate loosely-coupled solution algorithms for transient nonlinear computational aeroelasticity, *Comput. Methods Appl. Mech. Engrg.* 195 (17) (2006) 1973–2001, <http://dx.doi.org/10.1016/j.cma.2004.11.031>, URL <https://www.sciencedirect.com/science/article/pii/S0045782505001945> Fluid-Structure Interaction.
- [57] G. Zhang, S. Wang, Z. Sui, L. Sun, Z. Zhang, Z. Zong, Coupling of SPH with smoothed point interpolation method for violent fluid–structure interaction problems, *Eng. Anal. Bound. Elem.* 103 (2019) 1–10, <http://dx.doi.org/10.1016/j.enganabound.2019.02.010>, URL <https://www.sciencedirect.com/science/article/pii/S0955799718306295>.
- [58] C. Habchi, S. Russell, D. Bougeard, J.-L. Harion, T. Lemenand, A. Ghanem, D.D. Valle, H. Peerhossaini, Partitioned solver for strongly coupled fluid structure interaction, *Comput. & Fluids* 71 (2013) 306–319, <http://dx.doi.org/10.1016/j.compfluid.2012.11.004>, URL <https://www.sciencedirect.com/science/article/pii/S0045793012004367>.

- [59] Z.-Q. Zhang, J. Yao, G. Liu, An immersed smoothed finite element method for fluid-structure interaction problems, *Int. J. Comput. Methods* 8 (2012) 747–757.
- [60] M. Uhlmann, *New Results on the Simulation of Particulate Flows*, Technical Report No. 1038, 2004.
- [61] J. Stijnen, J. de Hart, P. Bovendeerd, F. van de Vosse, Evaluation of a fictitious domain method for predicting dynamic response of mechanical heart valves, *J. Fluids Struct.* 19 (6) (2004) 835–850, <http://dx.doi.org/10.1016/j.jfluidstructs.2004.04.007>, URL <https://www.sciencedirect.com/science/article/pii/S088997460400060X>.
- [62] S. Turek, J. Hron, Proposal for numerical benchmarking of fluid-structure interaction between an elastic object and laminar incompressible flow, in: H.-J. Bungartz, M. Schäfer (Eds.), *Fluid-Structure Interaction*, Springer Berlin Heidelberg, Berlin, Heidelberg, 2006, pp. 371–385.
- [63] E.H. R. Nemer, *Adaptive Immersed Mesh Method (AIMM) for Fluid Structure Interaction*, Elsevier, 2022.
- [64] M. Luhar, H.M. Nepf, Flow-induced reconfiguration of buoyant and flexible aquatic vegetation, *Limnol. Oceanogr.* 56 (6) (2011) 2003–2017, <http://dx.doi.org/10.4319/lo.2011.56.6.2003>, arXiv:<https://aslopubs.onlinelibrary.wiley.com/doi/pdf/10.4319/lo.2011.56.6.2003> URL <https://aslopubs.onlinelibrary.wiley.com/doi/abs/10.4319/lo.2011.56.6.2003>.
- [65] F.-B. Tian, H. Dai, H. Luo, J. Doyle, B. Rousseau, Fluid–structure interaction involving large deformations: 3D simulations and applications to biological systems, *J. Comput. Phys.* 258 (2014) 451–469, <http://dx.doi.org/10.1016/j.jcp.2013.10.047>.
- [66] E. Leveque, F. Toschi, L. Shao, J. Bertoglio, Shear-improved smagorinsky model for large-eddy simulation of wall-bounded turbulent flows, *J. Fluid Mech.* 570 (2007) 491–502, <http://dx.doi.org/10.1017/S0022112006003429>.
- [67] K. Zhou, S. Balachandar, An analysis of the spatio-temporal resolution of the immersed boundary method with direct forcing, *J. Comput. Phys.* 424 (2020) <http://dx.doi.org/10.1016/j.jcp.2020.109862>, URL <https://www.osti.gov/biblio/1850325>.